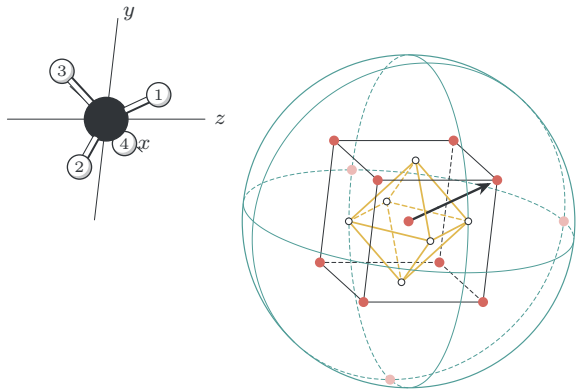
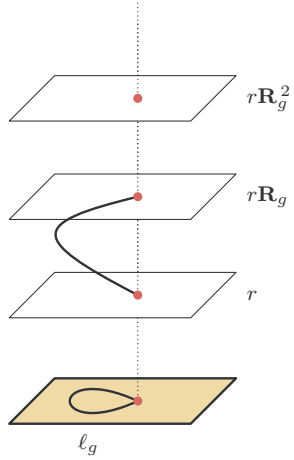


Methane,  $\text{CH}_4$ : the reference  $\mathbf{X}_0$  in its body frame,  
 with the  $C_2$  axes  $x, y, z$  through the midpoints of opposite edges.



The orientation space  $SO(3)$  as a ball,  
 direction the axis and distance the angle:  
 the twelve rotations of  $T$ ,  
 the cell  $\mathcal{U}$  at  $q = 0$  in gold,  
 and the lift  $r \rightarrow r\mathbf{R}_g$  of the third-turn loop.



The covering  $SO(3) \rightarrow SO(3)/T$ ,  
 as in Figure 11: three of the twelve  
 sheets, one per rotation in  $T$ ;  
 the loop  $\ell_g$  in the base, and its lift  
 from sheet  $r$  to  $r\mathbf{R}_g$ .  
 A sees 1,  ${}^1\text{E}$  sees  $\omega$ ,  ${}^2\text{E}$  sees  $\omega^2$ ,  
 and T cycles  $x, y, z$ .